

An exact growth constant for the quadratic polynomial continued fraction

(elementary, modulo one standard Gamma-ratio asymptotic)

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Abstract

For integers $A, B, C \geq 1$ let $b_k = Ak^2 + Bk + C$ and let q_n be the standard continuant denominators of the polynomial continued fraction $V(A, B, C) = 1 + \mathbf{K}_{n \geq 1} 1/(An^2 + Bn + C)$, i.e. $q_0 = 1$, $q_1 = b_1$, $q_n = b_n q_{n-1} + q_{n-2}$. We prove

$$\log q_n = n \log A + 2 \log(n!) + \frac{B}{A} \log n + K(A, B, C) + O(1/n),$$

with the additive constant in closed-plus-transcendental form $K = K_\Gamma + \delta$, where $K_\Gamma = -\log(\Gamma(1-r_1)\Gamma(1-r_2))$ is the elementary naive-product constant (r_1, r_2 the roots of $Ax^2 + Bx + C$) and $\delta = \log R_\infty > 0$, $R_\infty = \lim_n q_n / \prod_{k=1}^n b_k$, is a strictly positive continued-fraction correction. The two pieces are independent: K_Γ comes from the product $\prod b_k$ alone, and δ from the $+q_{n-2}$ term. At $B = 0$ one has $K_\Gamma(A, 0, C) = \log(\sinh(\pi\sqrt{C/A})/(\pi\sqrt{C/A}))$. This pins the additive constant left undetermined by the deposited denominator-growth bound $\log Q_n \geq n \log n - O(n)$ [1], sharpening that one-sided estimate (which suffices for irrationality by Euler’s criterion) to an exact two-sided expansion through $O(1/n)$ (Corollary 1); the elementary part K_Γ is closed-form, while $\delta = \log R_\infty$ is a strictly positive correction for which we claim *no* closed form. Every algebraic and asymptotic step below is independently confirmed numerically (`verify_structural.py`, `mpmath`, 50–60 digits).

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1 Introduction and relation to the deposited bound

The polynomial continued fraction $V(A, B, C) = 1 + \mathbf{K}_{n \geq 1} 1/(An^2 + Bn + C)$ has unit partial numerators, so its convergents p_n/q_n obey the Wronskian $|p_n q_{n-1} - p_{n-1} q_n| = 1$ and Euler’s irrationality criterion applies as soon as the denominators q_n grow fast enough. For that purpose one needs only a *one-sided lower bound* on $\log q_n$. The deposited catalogue [1] proves exactly such a bound: its denominator-growth lemma (`lem:Qgrowth`, “Denominator growth”) states, for $A \geq 1$, $C \geq 1$ and $An^2 + Bn + C > 0$ on $n \geq 1$,

$$\log Q_n \geq n \log n - O(n)$$

for the standard continuant $Q_n = (An^2 + Bn + C)Q_{n-1} + Q_{n-2}$ (its proof in fact yields the stronger $\log Q_n \geq 2n \log n - O(n)$), and uses it (via `prop:irrat`) to certify 482 quadratic constants irrational by Euler’s criterion. The *exact* size of $\log q_n$ — in particular its additive constant — is neither determined nor needed there.

This note determines it. For integers $A, B, C \geq 1$ we prove a two-sided asymptotic (Theorem 1) with an explicit additive constant $K = K_\Gamma + \delta$. The leading term $2 \log(n!) = 2n \log n - 2n + O(\log n)$ reproduces the deposited bound (Corollary 1), so the result is consistent with, and strictly sharper than, [1]: a one-sided estimate of the leading order is upgraded to a full expansion through $O(1/n)$ with the constant pinned. The constant separates cleanly into an elementary product part K_Γ (the constant of the naive product $\prod b_k$) and a continued-fraction correction δ (the $+q_{n-2}$ term); the two are genuinely distinct, since $q_n / \prod_{k \leq n} b_k \rightarrow R_\infty > 1$.

Two caveats fix the scope honestly. (i) The exact law is proved for integers $A, B, C \geq 1$, where both roots of $Ax^2 + Bx + C$ have negative real part; the deposited bound covers the slightly wider sign range $An^2 + Bn + C > 0$. (ii) While K_Γ is closed-form (at $B = 0$ it is the sinh expression above), we give *no* closed form for the correction $\delta = \log R_\infty > 0$: an 80-digit integer-relation (PSLQ) search against $\{1, \log 2, \log 3, \log \pi, \log(\sinh / \cdot), \gamma, \log \Gamma(\frac{1}{4})\}$ finds no relation *for the single triple* $(A, B, C) = (1, 0, 1)$ *against this basis* — which we record as suggestive evidence, not proof, that δ is non-elementary. The contribution is therefore the exact *structure* of the constant and the proof that the correction is strictly positive, not a closed form for δ .

Standing assumptions and well-posedness. Throughout A, B, C are integers ≥ 1 . The roots of $Ax^2 + Bx + C$ satisfy $r_1 + r_2 = -B/A < 0$ and $r_1 r_2 = C/A > 0$, so either they are a complex-conjugate pair or two negative reals; in all cases $\operatorname{Re} r_i < 0$, hence $1 - r_i$ avoids the non-positive integers and $\Gamma(1 - r_1)\Gamma(1 - r_2)$ is finite and, being either $|\Gamma(1 - r_1)|^2$ (conjugate roots) or a product of two positive reals (negative real roots), is real and positive. Thus K_Γ is real and well-defined. Also $b_k = Ak^2 + Bk + C \geq A + B + C \geq 3 > 0$ for all $k \geq 1$, so $q_n > 0$ and $P_n := \prod_{k=1}^n b_k > 0$. All Gamma arguments below lie in the open right half-plane $\{\operatorname{Re} z > 0\}$ (since $\operatorname{Re}(n + 1 - r_i) > 0$ and $\operatorname{Re}(1 - r_i) > 0$), where Γ is zero- and pole-free; we write $\operatorname{Log} \Gamma$ for the principal branch there, compatible with conjugation, $\operatorname{Log} \Gamma(\bar{z}) = \overline{\operatorname{Log} \Gamma(z)}$, so all complex Gamma logarithms below are unambiguous.

2 The product constant K_Γ

Lemma 1 (Exact product/Gamma identity). *For all $n \geq 0$,*

$$P_n = \prod_{k=1}^n (Ak^2 + Bk + C) = A^n \frac{\Gamma(n+1-r_1)\Gamma(n+1-r_2)}{\Gamma(1-r_1)\Gamma(1-r_2)}.$$

Proof. Factor $Ak^2 + Bk + C = A(k - r_1)(k - r_2)$, so $P_n = A^n \prod_{k=1}^n (k - r_1) \prod_{k=1}^n (k - r_2)$. For any r not a positive integer, the telescoping identity $\prod_{k=1}^n (k - r) = (1 - r)(2 - r) \cdots (n - r) = \Gamma(n + 1 - r)/\Gamma(1 - r)$ holds (immediate from $\Gamma(z + 1) = z\Gamma(z)$). Applying it to r_1, r_2 gives the claim. The right side is real because r_1, r_2 are real or conjugate. $\square$

Lemma 2 (Gamma ratio asymptotic). *For fixed r with $\operatorname{Re} r < 0$, as $n \rightarrow \infty$, with the principal branch $\operatorname{Log} \Gamma$ on $\{\operatorname{Re} z > 0\}$,*

$$\operatorname{Log} \Gamma(n + 1 - r) = \operatorname{Log} \Gamma(n + 1) - r \log n + \frac{r(r-1)}{2n} + O(1/n^2).$$

Proof. The standard ratio expansion $\Gamma(z + a)/\Gamma(z + b) = z^{a-b} \left(1 + \frac{(a-b)(a+b-1)}{2z} + O(1/z^2)\right)^1$ with $z = n + 1$, $a = -r$, $b = 0$ gives $\Gamma(n + 1 - r)/\Gamma(n + 1) = (n + 1)^{-r} \left(1 + \frac{r(r+1)}{2(n+1)} + O(1/n^2)\right)$. Taking

¹DLMF §5.11(iii) (*Ratios*), eq. 5.11.13, $\Gamma(z + a)/\Gamma(z + b) \sim z^{a-b} \sum_{k \geq 0} G_k(a, b) z^{-k}$ with $G_0 = 1$ and, from

logarithms and using $\log(n+1) = \log n + 1/n + O(1/n^2)$ and $\log(1 + \frac{r(r+1)}{2(n+1)}) = \frac{r(r+1)}{2n} + O(1/n^2)$,

$$\log \frac{\Gamma(n+1-r)}{\Gamma(n+1)} = -r \log n - \frac{r}{n} + \frac{r(r+1)}{2n} + O(1/n^2) = -r \log n + \frac{r(r-1)}{2n} + O(1/n^2). \quad \square$$

Lemma 3 (Product constant). *As $n \rightarrow \infty$,*

$$\log P_n = n \log A + 2 \log(n!) + \frac{B}{A} \log n + K_\Gamma + \frac{\kappa}{n} + O(1/n^2),$$

where $K_\Gamma = -\log(\Gamma(1-r_1)\Gamma(1-r_2))$ and $\kappa = \frac{1}{2}(r_1(r_1-1) + r_2(r_2-1)) = \frac{1}{2}(\frac{B^2}{A^2} + \frac{B}{A} - \frac{2C}{A})$.

Proof. Since $\log P_n$ is real (Lemma 1, $P_n > 0$), take its value as Log of the positive real P_n and expand each factor by Lemma 2. For conjugate roots the two contributions are complex conjugates, so $\text{Log } \Gamma(n+1-r_1) + \text{Log } \Gamma(n+1-r_2) = 2 \text{Re Log } \Gamma(n+1-r_1)$ and likewise for the $\Gamma(1-r_i)$ pair; for real roots every term is already real. Either way no $2\pi i$ ambiguity arises, and

$$\log P_n = n \log A + 2 \text{Log } \Gamma(n+1) - (r_1+r_2) \log n + \frac{r_1(r_1-1) + r_2(r_2-1)}{2n} - \log(\Gamma(1-r_1)\Gamma(1-r_2)) + O(1/n^2).$$

Now $\text{Log } \Gamma(n+1) = \log(n!)$ and $-(r_1+r_2) = B/A$. The closed form for κ follows from $r_1+r_2 = -B/A$, $r_1 r_2 = C/A$, whence $r_1^2 + r_2^2 = (r_1+r_2)^2 - 2r_1 r_2 = B^2/A^2 - 2C/A$ and $r_1(r_1-1) + r_2(r_2-1) = B^2/A^2 - 2C/A + B/A$. The sum $\frac{r_1(r_1-1) + r_2(r_2-1)}{2} = \kappa$ is real (symmetric in the conjugate pair). $\square$

3 The continued-fraction correction δ

Let $R_n := q_n/P_n$ (so $R_0 = R_1 = 1$).

Lemma 4 (Scaled recurrence). *For all $n \geq 2$, $R_n = R_{n-1} + \frac{R_{n-2}}{b_{n-1}b_n}$. Consequently (R_n) is strictly increasing, bounded above, and converges to a limit $R_\infty \in (1, \infty)$ with $R_\infty - R_n = O(1/n^3)$.*

Proof. Divide $q_n = b_n q_{n-1} + q_{n-2}$ by $P_n = b_n P_{n-1} = b_n b_{n-1} P_{n-2}$:

$$R_n = \frac{q_n}{P_n} = \frac{q_{n-1}}{P_{n-1}} + \frac{q_{n-2}}{b_{n-1}b_n P_{n-2}} = R_{n-1} + \frac{R_{n-2}}{b_{n-1}b_n}.$$

All $b_k > 0$ and $R_0, R_1 > 0$, so by induction $R_n > 0$ and the increment $R_n - R_{n-1} = R_{n-2}/(b_{n-1}b_n) > 0$; hence (R_n) is strictly increasing, and $R_2 = 1 + 1/(b_1 b_2) > 1$. By monotonicity $R_{n-2} \leq R_{n-1}$, so $R_n \leq R_{n-1}(1 + \frac{1}{b_{n-1}b_n})$, giving $R_n \leq \prod_{k=2}^\infty (1 + \frac{1}{b_{k-1}b_k})$. Since $b_{k-1}b_k \geq A^2(k-1)^2 k^2$, the sum $\sum_k 1/(b_{k-1}b_k)$ converges ($= O(\sum k^{-4})$), so the product converges and (R_n) is bounded. A monotone bounded sequence converges; the limit R_∞ satisfies $R_\infty \geq R_2 > 1$. Finally, since $b_{m-1}b_m \geq A^2(m-1)^2 m^2 \geq A^2(m-1)^4$ we have, by comparison with $\int_n^\infty t^{-4} dt$, $\sum_{m>n} 1/(b_{m-1}b_m) \leq A^{-2} \sum_{m>n} (m-1)^{-4} = O(1/n^3)$. Summing the (positive) increments past n ,

$$0 < R_\infty - R_n = \sum_{m>n} \frac{R_{m-2}}{b_{m-1}b_m} \leq R_\infty \sum_{m>n} \frac{1}{b_{m-1}b_m} = O(1/n^3). \quad \square$$

Lemma 5 (Correction constant). $\log R_n = \delta + O(1/n^3)$ with $\delta := \log R_\infty > 0$.

Proof. By Lemma 4, $R_n = R_\infty(1 + O(1/n^3))$ with $R_\infty > 1$, so $\log R_n = \log R_\infty + \log(1 + O(1/n^3)) = \delta + O(1/n^3)$, and $\delta > 0$ because $R_\infty > 1$. $\square$

eq. 5.11.17, $G_1(a, b) = \frac{1}{2}(a-b)(a+b-1)$ (here $a = -r$, $b = 0$, so $G_1(-r, 0) = \frac{1}{2}r(r+1)$). Equation numbers verified against the current DLMF release (dlmf.nist.gov/5.11); the leading z^{a-b} term and the $r(r-1)/2 \log$ -coefficient are also confirmed numerically in `verify_structural.py` (step S2, two-point Richardson, matching to ≥ 6 digits across the test cases).

4 The growth law

Theorem 1. *With the standing assumptions (in particular $A, B, C \geq 1$, so both roots r_1, r_2 of $Ax^2 + Bx + C$ have negative real part and every Gamma argument below avoids the poles, making K_Γ real and finite), as $n \rightarrow \infty$,*

$$\log q_n = n \log A + 2 \log(n!) + \frac{B}{A} \log n + (K_\Gamma + \delta) + \frac{\kappa}{n} + O(1/n^2),$$

where $K_\Gamma = -\log(\Gamma(1 - r_1)\Gamma(1 - r_2))$, $\delta = \log R_\infty > 0$, and $\kappa = \frac{1}{2}(B^2/A^2 + B/A - 2C/A)$. In particular the additive constant is $K(A, B, C) = K_\Gamma + \delta$, and $\log q_n = n \log A + 2 \log(n!) + \frac{B}{A} \log n + K + O(1/n)$.

Proof. $\log q_n = \log P_n + \log R_n$. Substitute Lemma 3 (which carries the κ/n term, $O(1/n^2)$ remainder) and Lemma 5 (which contributes δ with $O(1/n^3) \subset O(1/n^2)$ remainder, and no $1/n$ term). Adding gives the stated expansion; the $1/n$ coefficient is κ from the product alone. $\square$

Corollary 1 (Recovery and sharpening of the deposited bound). *For integers $A, B, C \geq 1$, Theorem 1 gives $\log q_n = 2n \log n + O(n)$; in particular $\log q_n \geq n \log n$ for all sufficiently large n . Hence, on the sub-family $A, B, C \geq 1$, Theorem 1 implies the deposited denominator-growth bound $\log Q_n \geq n \log n - O(n)$ of [1] (with the same leading constant $2n \log n$), while additionally fixing every term down to $O(1/n)$ and supplying the exact additive constant $K_\Gamma + \delta$.*

Proof. By Stirling $2 \log(n!) = 2n \log n - 2n + O(\log n)$; also $n \log A = O(n)$, $\frac{B}{A} \log n = O(\log n)$, and $K_\Gamma + \delta + \kappa/n = O(1)$. Summing the expansion of Theorem 1 gives $\log q_n = 2n \log n + O(n)$. Since the $O(n)$ remainder is $o(n \log n)$, $\log q_n \geq n \log n$ eventually, which is the deposited inequality; Theorem 1 additionally fixes all lower-order terms. $\square$

Remark 1 (The $B = 0$ slice). *At $B = 0$ the $(B/A) \log n$ term vanishes and $r_{1,2} = \pm i\sqrt{C/A}$, so by the reflection/product formula $\Gamma(1 - iy)\Gamma(1 + iy) = \pi y / \sinh(\pi y)$ with $y = \sqrt{C/A}$, giving $K_\Gamma(A, 0, C) = \log(\sinh(\pi\sqrt{C/A})/(\pi\sqrt{C/A}))$. This is the elementary product constant; the true constant for q_n is $K_\Gamma + \delta$ with $\delta = \log R_\infty > 0$, i.e. strictly larger.*

Remark 2 (Relation to the deposited irrationality bound). *The deposited paper [1] uses the same standard continuant Q_n (e.g. for $A = C = 1, B = 0$ one has $Q_3 = 112$, not the product value 100), so $\delta = \log R_\infty$ is genuinely present in its object; but it establishes only the one-sided lower bound *lem:Qgrowth*, $\log Q_n \geq n \log n - O(n)$, which is all Euler’s criterion (*prop:irrat*) requires. By Corollary 1 Theorem 1 reproduces and strictly sharpens that bound on the $A, B, C \geq 1$ sub-family. It contradicts nothing deposited: no deposited manuscript pins the additive constant, and the only “sinh” elsewhere in the corpus is a Bessel-ratio value of one even continued fraction, unrelated to the present growth constant.*

Epistemic status

Four-class (SIARC) grading:

- **STRUCTURAL** (hand proof; not yet machine-checked in Lean): the full Theorem 1, including $K = K_\Gamma + \delta$, $\delta > 0$, the closed form for κ , and the $B = 0$ sinh identity. The argument is elementary (Lemmas 1–5) and complete; it is labelled STRUCTURAL rather than PROVEN solely because it has not been formalized.

- **VERIFIED** (numerically, finite cases): every step is confirmed by `verify_structural.py` (mpmath, 50–60 digits) over the seven triples

$$(A, B, C) \in \{(1, 0, 1), (2, 0, 1), (1, 0, 3), (1, 1, 1), (1, 3, 1), (3, 2, 5), (2, 3, 7)\} :$$

S1 the exact product/Gamma identity; S2 the $r(r-1)/2$ ratio coefficient (Richardson); S3 product constant $\rightarrow K_\Gamma$; S4 R_n monotone, bounded, $R_\infty > 1$, $O(1/n^3)$ tail; S5 full-law residual $\rightarrow \kappa/n$ against the closed form. The cross-check $K_\Gamma \stackrel{B=0}{=} \sinh$ holds to 50 digits.

- **CONJECTURED**: $\delta = \log R_\infty$ is conjectured to be non-elementary (no closed form). The evidence is a single null PSLQ run (at 80 digits, for the one triple $(1, 0, 1)$, against $\{1, \log 2, \log 3, \log \pi, \log(\sinh / \cdot), \gamma, \log \Gamma(1/4)\}$): no relation found. This is suggestive for that case and basis only, not a general theorem; a null PSLQ result is a complete, honest answer, not a gap to be filled.
- **Standard input (confirmed)**: the sole non-self-contained step is the Gamma-ratio asymptotic of Lemma 2 — DLMF §5.11(iii) eqs. 5.11.13 and 5.11.17, whose equation numbers were verified against the current DLMF release. The proof is otherwise elementary and self-contained, so the result is best described as “elementary modulo one standard cited asymptotic.”

The natural **PROVEN** upgrade is a Lean finitary core: Lemma 1 (an algebraic identity) and Lemma 4’s monotone–bounded convergence are directly formalizable; the Gamma/Stirling asymptotics can be taken as an explicit labelled hypothesis (the conditional-core pattern).

References

- [1] Papanokechi, *Polynomial Continued Fractions: a Proved Logarithmic Ladder, a $4/\pi$ Casoratian Identity, and 482 Irrational Constants*, 2026. Denominator-growth lemma: `Qgrowth` and irrationality proposition `prop:irrat`. Source `source.tex` in directory `polynomial-continued-fractions-proved-logarithmic-laddera4casoratian/` of the repository `papanokechi/submitted-manuscripts` at commit `c45e93b`.

*FLAG(operator): insert the deposited paper’s own (concept/version) DOI here once minted; do **not** reuse the M9 DOI 10.5281/zenodo.20542161, which is a different artifact.*

- [2] F. W. J. Olver et al. (eds.), *NIST Digital Library of Mathematical Functions*, <https://dlmf.nist.gov/>, §5.11(iii) (*Ratios*), eqs. 5.11.13 and 5.11.17 (equation numbers verified against the current release).

Code and data availability. The verifier `verify_structural.py` and harness `harness_quad.py`, reproducing every numerical check (steps S1–S5), are available in the repository `papanokechi/pcf-quadratic-growth` (<https://github.com/papanokechi/pcf-quadratic-growth>).

AI-assistance disclosure. Prepared with AI assistance (GitHub Copilot; Anthropic Claude) for drafting, computer-algebra verification, and manuscript preparation. The numerical verification was obtained by running `verify_structural.py` (mpmath, 50–60 digits); the author takes responsibility for the content and for the placement of each claim within the four-grade epistemic partition.